

Unmanned Aerial Systems Mission Task Element Development: Lessons Learned from Flight Test

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As unmanned aerial systems (UAS) become ever more prevalent and diverse in their application (control line inspection, package delivery, surveillance, etc.), their inclusion in the national airspace is unavoidable. New use cases for these vehicles are being formed every day, and the variety of their design forms (fixed wing, rotorcraft, quad-copters and greater, tail sitters, etc.) and sizes (0.55 to 10's of thousands of pounds) will prove challenging to certify. To meet the stringent requirements of both the government and public expectations of safety, a robust procedure must be established through which the vehicle's handling and flying qualities can be determined. In the realm of manned vehicles, this has been accomplished through the application of longstanding requirements on vehicle performance, dynamics, and controllability as defined by the Federal Aviation Regulations in the civilian world or standards, such as MIL-STD-1797, in the military domain. For military rotorcraft, a mission-oriented approach has been in usage for decades, as defined in ADS-33E-PRF, and is well accepted by the rotorcraft community. To address the issue of UAS certification, a team led by Systems Technology Inc. developed a process to guide the UAS community through the handling qualities evaluation of a UAS. Given the current and expected diversity in vehicle type, size and mission, the mission-oriented approach was selected for this application. A demonstration catalog of candidate mission task elements (MTE), specific tasks with well-defined evaluation processes and criteria, have been defined and flight tested using a representative UAS. The results of these evaluations are presented herein.

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I. Introduction

A. Traditional Classification Schemes and UAS Challenges

Military flying qualities specifications and levels for fixed-wing aircraft have traditionally been defined and distinguished by very specific characteristics: size, weight and vehicle flight phase. This has been the standard for decades, as defined in MIL-F-8785 and MIL-STD-1797 for manned aircraft. The vehicles have been categorized into four Classes, and the mission phases defined in three Categories. The four fixed-wing vehicle classes are: Class I) small, light air vehicles; Class II) low-to-medium maneuverability aircraft, such as search and rescue, light or medium cargo, tactical bomber, heavy attack; Class III) large and heavy vehicles, such as heavy transport or bomber; and Class IV) very maneuverable aircraft, such as a fighter, interceptor, etc. Category A includes rapid maneuvers and precise tracking or flight path control. Category B includes gradual maneuvers without a precision requirement. Category C includes terminal flight phases (takeoff, landing, etc.) and generally requires a gradual but accurate control strategy.

Challenges to this system become apparent if a vehicle performs the mission of a particular Class but does not look like or behave dynamically in the same manner as vehicles typical of that class. To use the example of Ref. [1], how does an engineer address the discrepancy between the AC-130 gunship and the C-130J transport? Both are Class III vehicles in the traditional system but with very different missions that require different capabilities of the vehicle. The AC-130 is an attack vehicle, similar in role to the A-10. Why should the requirements for the A-10, a Class IV vehicle, be stricter than those for the AC-130, or those for the AC-130 more lenient than those for the A-10, due to their airframe particulars when the mission they are meant to perform is the same?

In addition, autonomy poses new challenges to the assessment process, not addressed by the current standards. In days of old, only human pilots operated the vehicles, performed the assessments, rated and reviewed the vehicle through defined processes and scales, such as the Cooper-Harper [2] and Pilot-Induced Oscillation (PIO) Tendency scales [3,4]. Considerations for the inceptor dynamics and cockpit displays were a natural part of the evaluation. If poorly designed or implemented, they could adversely affect the capabilities and evaluation of the vehicle. An autonomous system has no need for those elements: the inceptor or displays do not play a role in the assessment process. Instead, functions, such as guidance, navigation and control, take on a more critical role and influence the ease of operation and controllability of the vehicle. A range of autonomy levels must also be considered. The vehicle may be strictly remotely piloted (requiring consideration of the displays and inceptor again), assisted by a remote pilot, fully autonomous, or some combination of the three.

Each of the above considerations is explored in more detail in Ref. [1]. A case for the application of the mission-oriented approach or MTE has been made in Ref. [1] as well. The purpose of the present paper is to provide additional background on the origin of MTE, highlight example MTE that have been flown and evaluated [5], and provide a set of lessons learned to guide the future design and application of the current and new MTE.

B. MTE Background and History

MTE have their origins in the rotorcraft world, formally first defined in the then newly formed rotorcraft handling qualities standard, ADS-33 [6]. The genesis of this new standard and the introduction of MTE began with the original rotorcraft flying and ground handling qualities requirements document that was released in 1958 and updated in 1961, MIL-H-8501A [7]. The document contained "...relatively simple time-response requirements on stability and control power and damping requirements stated in terms of weight and inertia" [8]. Note that the handling qualities component was strictly ground based and did not address any airborne requirements, the latter left in the domain of flying qualities.

Numerous follow-on studies of MIL-H-8501A revealed an absence of criteria or methods for addressing the highly coupled nature of nonlinearities present in helicopters [9]. Key in reference [10] identified a need for a specification structure similar to that in MIL-F-8785B [11], a systematic treatment of flying qualities, flight envelopes and reliability. Goldstein [12] cites a lack of handling qualities data, inadequate hover control power criteria, incomplete visual flight rules dynamics requirements guidance, deficient differentiation between hover and forward flight handling qualities characteristics, and limited guidance for degraded flying qualities.

A Vertical/Short Take Off and Landing specification, MIL-F-83300 [13], was published in 1970, based on research sponsored by the Air Force [8]. The specification included a Background Information Users Guide (BIUG) published in 1971 [14]. At the time, only the Air Force adopted this specification, with the Army and Navy

continuing to use MIL-H-8501 [8]. This was deemed inadequate for helicopters at that time as it still did not address the nuances of strong inter-axis couplings and nonlinearities, nor were its supporting data rotorcraft specific, as it had been collected for VSTOL vehicles [8].

Revisions continued to refine this document, resulting in the latest release, ADS-33E-PRF in 2000. An effort is currently ongoing to update this specification, again, to capture more recent research. ADS-33E-PRF is widely accepted as the rotorcraft handling qualities standard and has been applied by researchers, government, and industry. The MTE process and benefits are well understood within this community. Applications of the methodology have been expanding more recently to include UAS, as discussed in the remainder of this paper and elsewhere [1], Urban Air Mobility [15], and proposed FAA certification efforts for e-VTOL vehicles under a Broad Agency Announcement project undertaken by STI and others (“A Study of Envelope Protection Approaches & Advanced Flight Control Modes for Protection Against Loss of Control in Part 23 Airplanes”).

C. Proposed UAS MTE Structure

The currently proposed UAS MTE template is given below. The intent is to capture the traditional task objectives, description and performance requirements already contained in rotorcraft MTE, while also including notes on autonomy levels and relevant MTE development details. The autonomy aspect in particular will play a significant role in the UAS evaluations. The template includes notes on the suggested content for each section.

MTE Name (Precision/Aggressiveness Level)

Autonomy Level

- Specifies expected operator intervention.
- If MTE can be flown with various amounts of intervention, and any part of the MTE changes as a result, consider creating separate MTE for different levels of autonomy.

Task Objectives

- Two-to-four high-level bulleted items that will help the user determine why this MTE should be used, its applicability to the vehicle in question, and the expected outcomes.

Task Description

Brief and explicit description of the task, including test course layout and specialized equipment/displays needed. Keep the MTE simple in operation. Consider breaking elaborate tasks into two or more MTE. Be careful setting time as a task parameter. Consider whether time is a part of the task description (meaning it must be met) or a performance limit (meaning it is a measure of goodness).

Desired Performance

- Bulleted list of the expected levels of task performance for a UAS that is satisfactory without improvement.
- List those parameters you think are important, even if the limits are listed as “TBD” for now.

Adequate Performance

- Bulleted list of the expected levels of task performance for a UAS that has deficiencies that warrant improvement.
- List those parameters you think are important, even if the limits are listed as “TBD” for now.

Notes for developing this MTE:

1. Any additional comments are provided here. This could relate to recommended processes of conducting the MTE, potential pitfalls, and other considerations.

II. MTE Test Campaign

A. Background

In support of the MTE development process, a series of evaluation flights were conducted by the University of Minnesota (UMN) from November 2018 through March 2019 using an UltraStick120. Table 1 provides a summary of the test conditions for each evaluation flight. The purpose of the flights was to evaluate the MTE against a set of known vehicle configurations, to identify the strengths, weaknesses, and requirements of each. This is the first step in formally defining a final version of the MTE. More test flights will be required from a wide range of vehicle sizes and capabilities to improve the MTE definition. Earlier flights were conducted to perform system identification on the vehicle bare airframe dynamics and actuator models [16]. The identified characteristics are shown in this paper and represent the vehicle flown.

Table 1: UltraStick120 MTE Flight Test Run Log – Test Conditions

Flight	Flight Type	Speed	Condition		
12	MTE	23/17 m/s	Cruise/Approach	Normal CG	No Delay/Added Delay
13	MTE	23/17 m/s	Cruise/Approach	Normal CG	No Delay/Added Delay
14	MTE	23/17 m/s	Cruise/Approach	Normal CG	No Delay/Added Delay
16	MTE	23/17 m/s	Cruise/Approach	Normal CG	No Delay/Added Delay

The MTE flown in each flight are shown in Table 2. Detailed descriptions of each follow.

Table 2: UltraStick120 MTE Flight Test Run Log – MTE Flown

Flight	MTE Evaluated
12	a) Flightpath Regulation in the Presence of a Discrete Gust (1-Cosine excitation to pitch rate) b) Flightpath Regulation in the Presence of a Sum-of-Sines Disturbance (Non-Precision, Non-Aggressive (Altitude)) c) Precision Lateral Offset Landing
13	a) Flightpath Regulation in the Presence of a Discrete Gust (1-Cosine excitation to pitch rate) b) Flightpath Regulation in the Presence of a Sum-of-Sines Disturbance (Non-Precision, Non-Aggressive (Altitude)) c) Precision Lateral Offset Landing
14	a) Flightpath Regulation in the Presence of a Discrete Gust (1-Cosine excitation to pitch rate) b) Flightpath Regulation in the Presence of a Sum-of-Sines Disturbance (Non-Precision, Non-Aggressive (Altitude)) c) Waypoint following d) Precision Lateral Offset Landing
16	a) Flightpath Regulation in the Presence of a Sum-of-Sines Disturbance (Precision, Non-Aggressive (Attitude)) b) Waypoint following c) Precision Lateral Offset Landing

B. Test Vehicle

1. Hardware

The UAS laboratory at UMN maintains the UltraStick series of fixed-wing UAS to serve as testbeds in several ongoing research projects, such as control law design, navigation and guidance and fault detection, isolation and reconfiguration (Ref. [17]). The series consists of the UltraStick120, UltraStick25e and the UltraStick mini. The laboratory also supports UAS related research of other organizations by providing flight-testing services on these testbeds. The UltraStick120 flown for these evaluations is shown in Fig. 1, pictured in the center. While only the UltraStick120 results are reported here, this vehicle design is widely used in the sUAS industry and is representative of a broad category of vehicles.



Fig. 1: UltraStick120 (center) and UltraStick25e aircraft.

The UltraStick120 was a commercially available, RC hobbyist airframe. The UMN UAS Lab converts such airframes into UAS platforms for research. Relevant size and performance values for the UltraStick120 are shown in Table 3.

Table 3: UltraStick120 Aircraft Parameters

Wingspan	1.92 m
Chord	0.43 m
Length	1.73 m
Airframe Weight	6.0 kg
Max Weight	10.0 kg
Cruise speed	23 m/s (typical)
Stall speed at Max Weight	13 m/s
Airspeed range	10-41 m/s

The aircraft is equipped with a commercial RC hobbyist control system (propulsion system, batteries, receiver, and servos) for manual control, as well as additional systems for data collection and autonomous flight modes. The implemented system-level design allows the pilot to take manual control of the aircraft. In addition, the manual control system failsafe defaults to a descending, power-off spiral to keep the aircraft contained within the test range.

The aircraft is powered by an electric motor and has removable wings for transport. The UltraStick120 has a conventional horizontal and vertical tail with rudder and elevator control surfaces. The aircraft has a symmetric airfoil wing with aileron and flap control surfaces. The rudder and elevator are actuated by Hitec HS5245MG servos. The flaps and ailerons use Hitec HS5625MG servos. The aircraft is propelled by a 1900W Actro 40-4 brushless electric motor with a Graupner 14 x 9.5 folding propeller. Power for the motor comes from two 5000mAh 5-cell lithium polymer batteries connected in series. The servos are powered by a separate 1350 mAh 3-cell lithium polymer battery. The main internal payload bay is located in the fuselage, directly under the wing; additional payloads may be accommodated in the aft fuselage or externally. The UltraStick120 aircraft has space aft of the avionics bay for additional sensors or payloads. SUAS with these general specifications and configuration are typical of the sUAS community at large.

2. Aircraft Bare Airframe Dynamics, Controller and Actuators

The aircraft controller implements a simple pitch and bank attitude tracker. The pitch attitude tracker is shown in Fig. 2. The internals of the Pitch Tracker block are shown in Fig. 3. The Roll Tracker block is an exact duplicate, but with unique gains. A yaw damper was included as a simple washout filter in the yaw rate to rudder feedback loop. As flown with the UltraStick120, the pitch axis features an attitude command system with speed hold, while the roll axis featured an attitude command, and the yaw axis features a rate command system.

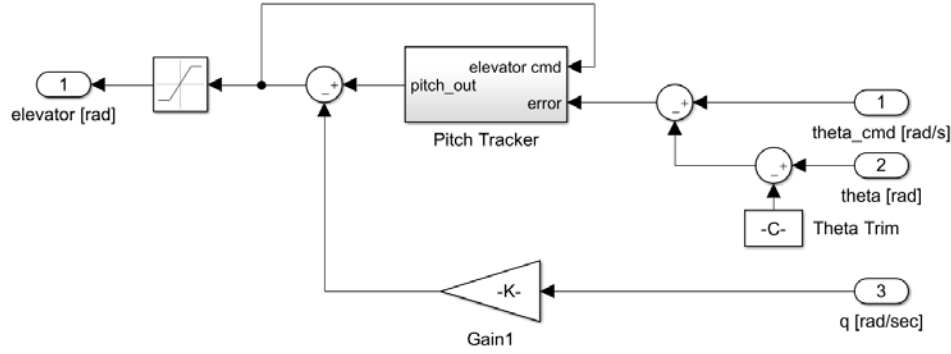


Fig. 2: Pitch attitude tracker.

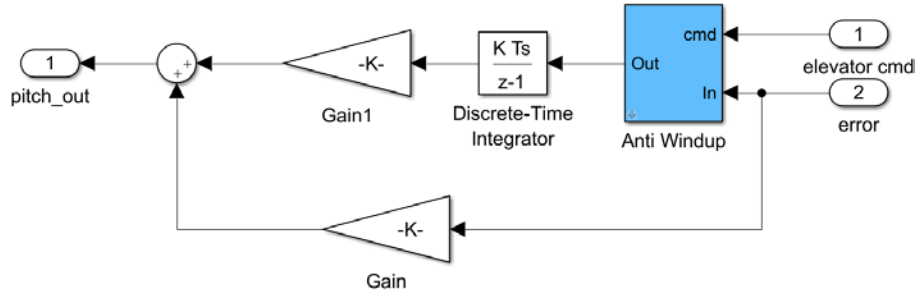


Fig. 3: Representative tracker block.

The aircraft actuators were modeled with a simple first-order model and delay, provided below. In addition, the bare airframe dynamics of q/δ_e are provided as well.

$$\frac{\delta_e}{\delta_{ec}} = \frac{8}{s+10} e^{-0.065s}, \quad \frac{q}{\delta_e} = \frac{40.54(0)(0.3175)(4.617)}{[0.1748, 0.6876][0.6664, 8.077]}$$

C. Test Facilities

The flight test facilities, shown in Fig. 4, consist of the UAV laboratory on the UMN campus and the airfield located at the UMore Park Test Range. The aircraft maintenance and pre-flight checkouts (simulation-in-the-loop, hardware-in-the-loop, etc.) occur in the UAV laboratory. The flight tests took place at the UMore Park Test Range near Rosemount, MN. The Test Range is located on sparsely populated agricultural fields owned by UMN. Flights are conducted year-round. A winter flight is shown in Fig. 4.

The UMore Park Test Range is located within Class G airspace. The test aircraft must remain within the area centered at 44° 43'32.71"N and 93° 4'44.49"W, with a radius of 0.28 NM and a ceiling of 400 feet Above Ground Level (AGL). All flights are coordinated with the Rosemount Research and Outreach Center Manager. In addition, the UMN UAV lab has obtained the proper Certificates of Authorization (COAs) for legal operation of UAS flights at the UMore Test Range. Non-participants are kept clear of this zone during operations. The dimensions comply with the 1-mile lateral and 400 foot ceiling limits and were created to avoid structures and major roadways while staying within direct line of sight of the operations crew. Restriction to the flight research zone minimizes risk to non-participants. The Operations Lead, Pilot, and Observer ensure that the aircraft remains within the flight research zone and will abort the mission if there is a risk of exiting the area. Failsafes on the aircraft ensure that the aircraft will remain in the test area in the event of link loss. Launch and recovery are conducted from a 900 foot by 60 foot turf runway aligned in the East-West direction. A right pattern is used when launching/recovering in the east-bound direction and a left pattern is used when launching/recovering in the west-bound direction. These patterns are used to avoid overflying the operations crew.



a) Laboratory



b) Airfield

Fig. 4: UAS flight test facilities at the University of Minnesota.

D. Candidate Mission Task Elements

Four distinct MTE are described below. The values are notional, used as a part of the initial test campaign. One set of tests with one vehicle is not exhaustive. However, the intention of this test was to exercise the MTE and adjust them based on the lessons learned, rather than define the final requirements.

1. *Flightpath Regulation in the presence of a Discrete Gust (Non-precision, Non-aggressive or Non-Precision, Aggressive, depending on gust input magnitude)*

Autonomy Level

- MTE can be applied to any level of autonomy.
- Intended to be a “surprise” disturbance, so the pilot should not be alerted if under manual control. No assumptions required for delayed pilot response.

Task Objectives

- Assess response to and recovery from a discrete gust.
- Evaluate vehicle control following a perturbation in pitch or flight path.

Task Description

In straight, level flight at constant airspeed, apply a gust to perturb the aircraft vertically. After the aircraft has recovered from the gust, apply a gust of equal amplitude and opposite sign. Assess the ability to reject the gust and return to equilibrium flight at the initial altitude and airspeed.

No specific ground-referenced course is needed. The MTE may be accomplished at any initial altitude and airspeed. With the aircraft trimmed for level, unaccelerated flight, apply the gust documented below in either the up or down direction; once the aircraft has recovered from the gust (defined as meeting either the Adequate or Desired performance standards), apply an equal and opposite gust. The maneuver is complete when the aircraft has returned to equilibrium flight.

The gust has a “1 – cosine” shape as shown in Fig. 5. The gust shall be applied to the most direct controller for vertical flight path (e.g., elevator for a conventional UA or rotor RPM/collective for a VTOL UAS), with magnitude v_m selected to provide an altitude change of at least 10 ft in one second after application, determined for the UA with outer-loop hold modes (or pilot intervention, for an RPA) inactive. The time d_m shall be tuned to the estimated rigid-body short-term response mode of interest. It is acceptable to replace the “1 – cosine” shape with a discrete gust.

The “1 – cosine” gust has the following shape:

$$\begin{aligned}
 v &= 0 & \text{for } x < 0 \\
 v &= v_m [1 - \cos(\pi x/d_m)]/2 & \text{for } 0 \leq x \leq d_m \\
 v &= v_m & \text{for } x > d_m
 \end{aligned}$$

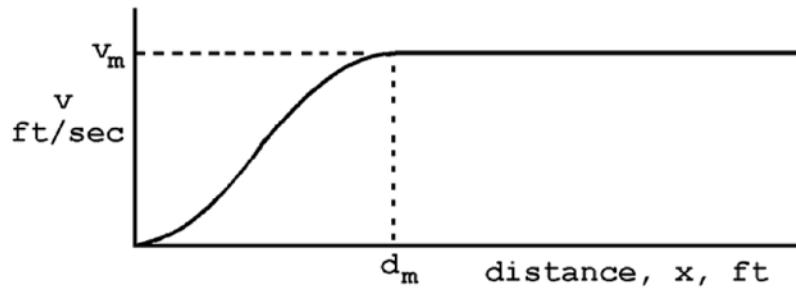


Fig. 5: “1 – cosine” shape for discrete gust.

Desired Performance

- Maximum altitude deviation following gust, ± 20 ft.
- Recover to initial altitude (for height hold systems) within ± 2 ft.
- Recover to initial trim attitude (for attitude hold systems) within ± 1 deg.
- Recover airspeed to initial trim conditions within ± 2 kt.
- Hold trim conditions following recovery for at least 10 sec.
- Retain initial heading within ± 1 deg.
- Time to recover to within performance standards 10 sec.

Adequate Performance

- Maximum altitude deviation following gust, ± 30 ft.
- Recover to initial altitude (for height hold systems) within ± 5 ft.
- Recover to initial trim attitude (for attitude hold systems) within ± 2 deg.
- Recover airspeed to initial trim conditions within $\pm 10\%$ of trim.
- Hold trim conditions following recovery for at least 5 sec.
- Retain initial heading within ± 2 deg.
- Time to recover to within performance standards 15 sec.

Notes for developing this MTE:

1. The input is based on MIL-STD and FAA gust requirements, but the proper values for UAS are entirely unknown. Suggest running gusts in basic model (no autopilot or manual reactions) varying d_m and v_m including pure discrete gusts.
2. This could be viewed to be equivalent to a small UAS flying over a low building or changing terrain: how well can the UAS handle the upset? Is it trivial, or will the UAS get blown into power lines, a tree, etc.?
3. Suggest the first gust be up and the second down; initially probably can get by with just the initial up gust, until there is confidence the down gust is not a safety issue.
4. Want to measure aircraft flight path and pilot control inputs to correct for the gust, with the pilot not knowing the gust is coming; maybe after a few runs with the pilot knowing it is coming, try a few with random application.

2. Flightpath Regulation in the presence of a Sum-of-Sines Disturbance (Precision, Aggressive)

Autonomy Level

- MTE can be applied to any level of autonomy.
- Intended to be a continuous, “random” disturbance that the pilot/autonomous controller must regulate against to maintain flightpath. No assumptions required for delayed pilot response.

Task Objectives

- Assess ability of pilot/autonomous controller to maintain flightpath in the presence of a “random” appearing continuous disturbance.
- Evaluate vehicle control following a continuous perturbation in pitch or flight path.

Task Description

In straight, level flight at constant airspeed, apply a sum-of-sines disturbance input to perturb the aircraft vertically. Continuously regulate desired flightpath in the presence of the disturbance for a minimum of 20 seconds. Assess the ability to reject the disturbance and maintain flightpath at the initial airspeed.

No specific ground-referenced course is needed. The MTE may be accomplished at any initial altitude and airspeed. With the aircraft trimmed for level, unaccelerated flight, apply the disturbance documented below. The maneuver is complete when the aircraft has responded to the continuous disturbance for at least 20 seconds and has returned to equilibrium flight.

The sum-of-sines (SOS) disturbance forcing function is used to drive the flightpath regulation task through which the pilot or autonomous controller attempts to minimize the flightpath error within desired performance constraints. Table 4 presents the parameters for a Fibonacci series-based SOS input that has been designed for the anticipated longitudinal dynamics of a typical fixed wing sUAS. The input is defined for a 20 second scoring time run length. Thus, each sine wave frequency is defined by f_n (Hz) = N_n (cycles/run)/20 (s/run). An example disturbance signal with a gain of 1.5 deg placed on the nominal signal is shown in Fig. 6.

Table 4: Example SOS Input Forcing Function Parameters for Lower Frequency Identification.

Frequency No.	1	2	3	4	5	6	7
Cycles/Run, N_n	3	5	8	13	21	34	55
Frequency, f_n (Hz)	0.1500	0.2500	0.4000	0.6500	1.0500	1.7000	2.7500
Frequency, (r/s)	0.9425	1.5708	2.5133	4.0841	6.5973	10.6814	17.2788
Amplitude ($A_i = f_i/f_n$)	+1.0000	-0.6000	+0.3750	-0.2308	+0.1429	-0.0882	+0.0545
Initial $A_n \times \omega_n$	Rate						
	+0.9425	-0.9425	+0.9425	-0.9425	+0.9425	-0.9425	+0.9425

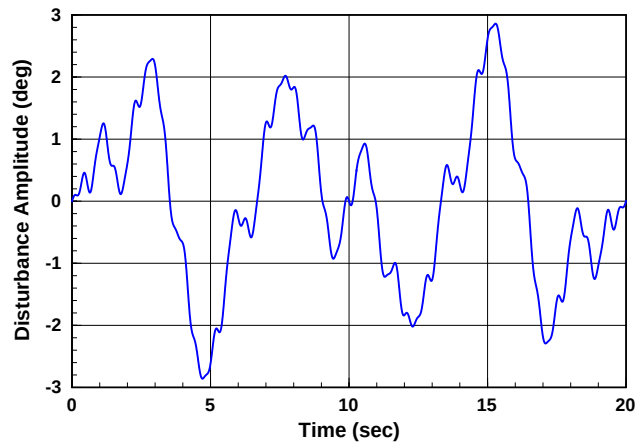


Fig. 6: Sum-of-sines disturbance (gain = 1.5 deg)

Desired Performance (Precision Task – Pitch Disturbance)

- Maintain flightpath during disturbance within ± 1 deg for at least 50% of the scoring time.
- Maintain altitude during disturbance through the disturbance, ± 20 ft.
- Maintain airspeed during disturbance to initial trim conditions within ± 2 kt.
- Maintain initial heading during disturbance within ± 1 deg.

Adequate Performance (Precision Task – Pitch Disturbance)

- Maintain flightpath during disturbance within ± 2 deg for at least 75% of the scoring time.
- Maintain altitude during disturbance through the disturbance, ± 40 ft.

- Maintain airspeed during disturbance to initial trim conditions within ± 5 kt.
- Maintain initial heading during disturbance within ± 2 deg.

Desired Performance (Non-Precision Task – Altitude Disturbance)

- Maintain altitude during disturbance through the disturbance, ± 10 ft.
- Maintain airspeed during disturbance to initial trim conditions within ± 2 kt.
- Maintain initial heading during disturbance within ± 1 deg.

Adequate Performance (Non-Precision Task – Altitude Disturbance)

- Maintain altitude during disturbance through the disturbance, ± 20 ft.
- Maintain airspeed during disturbance to initial trim conditions within ± 5 kt.
- Maintain initial heading during disturbance within ± 2 deg.

Notes for developing this MTE:

1. Gain on SoS disturbance and performance requirements will be refined as part of the flight test evaluations.
3. *Waypoint Following (Non-Precision, Non-Aggressive and Precision, Non-Aggressive, depending on task requirements)*

Autonomy Level

- Intended to assess autonomous operations.

Task Objectives

- Assess ability of an autonomous controller to follow a waypoint flightpath.

Task Description

In straight, level flight at constant airspeed, initiate a waypoint following autonomous control command sequence. Waypoint legs will feature a minimum of 500 ft (or approximately 150 m). The waypoint sequence will feature at least two turns about a rectangular/square path with an altitude change of at least 50 ft on one of the legs.

Desired Performance (Precision Task)

- Maintain lateral, longitudinal, and vertical flightpath position within ± 2 ft of the defined waypoint path within each leg.
- No undesirable control motions.

Adequate Performance (Precision Task)

- Maintain lateral, longitudinal, and vertical flightpath position within ± 5 ft of the defined waypoint path within each leg.
- No undesirable control oscillations.

Desired Performance (Non-Precision Task)

- Maintain lateral, longitudinal, and vertical flightpath position within ± 10 ft of the defined waypoint path within each leg.
- No undesirable control motions.

Adequate Performance (Non-Precision Task)

- Maintain lateral, longitudinal, and vertical flightpath position within ± 25 ft of the defined waypoint path within each leg.
- No undesirable control oscillations.

Notes for developing this MTE:

1. Specific waypoints and performance requirements will be refined as part of the flight test evaluations.
4. *Precision Lateral Offset Landing (Precision, Non-Aggressive)*

Autonomy Level

- Typically, this will be a remotely piloted MTE but may be flown manually.
- No assumptions required for delayed pilot response. Any response delays will be reflected in MTE performance.

Task Objectives

- Evaluate ability to precisely control horizontal and vertical flightpath and airspeed.
- Evaluate ability to precisely control sink rate and attitude in the flare.
- Evaluate tendency for nose bobble or PIO.
- Evaluate control sensitivity and harmony in landing.

Task Description

The offset landing task consists of a visual approach, during which the evaluation pilot (EP) aligns the aircraft approximately 50 feet off the runway centerline (see Fig. 7 for an example setup). At approximately 65 ft AGL, the EP corrects back to the centerline and attempts to touchdown within the desired parameters. Offsets to the left or right can be used interchangeably; however, the direction of offset may be dictated by pilot's line-of-sight.

A Precision Offset Landing test course at the UMore Park Test Range is shown in Fig. 7. In the figure, the yellow area is the usable turf runway (approximately 900 x 100 ft). An extended centerline in light blue has been added to the figure. The south edge of the runway adjoins a crop field with a distinct edge. The notional “base” and “final” shown in light green will position the aircraft along that edge. The remote pilot will be positioned along that line in order provide a good perspective for tracking during the turn to final, and during the final approach. The offset start will be based on altitude. Temporary markings for the runway centerline and “touchdown” target markers will be made. A landing “T” and an extended centerline can be painted on the turf.



Fig. 7: Precision offset landing course at the UMore Park test range.

Desired Performance

- Approach airspeed maintained within ± 2 kts.
- Touchdown within 5 feet of centerline (nose wheel on centerline).
- Touchdown within ± 5 feet of aimpoint.
- Sink rate – smooth touchdown.
- No PIO.

Adequate Performance

- Approach airspeed maintained within -2 kts/+5 kts.
- Touchdown within 10 feet of centerline.
- Touchdown within ± 10 feet of aimpoint.
- No PIO.

Notes for developing this MTE:

1. Correction height (altitude) and performance requirements will be refined as MTE is initially evaluated.

III. Results

The following contains a summary of the notes, insights, and evaluations of the candidate MTE.

A. Flightpath Regulation Tasks

1. Flightpath Regulation in the Presence of a Discrete Gust:

An example flightpath regulation task is shown in Fig. 8. The tasks were flown in a “pseudo” autonomous mode, where the remote pilot placed the vehicle on condition, while the on-board controller responded to the disturbance. The aggressiveness of the task can be adjusted based on the size of the input. For this example, the aggressiveness level was set by a 1-Cosine gust disturbance, based on a 2 rad/s pitch rate disturbance for the 23 m/s cruise flight

condition. Note that a turn is initiated just prior to the 1370 second time index as seen in the heading response (Fig. 8d). Desired and adequate performance requirements are indicated by the green and yellow bands, respectively. Prior to the turn, the vehicle maintains airspeed and altitude within desired tolerances. However, the pitch attitude response often falls out of the adequate bounds. The UltraStick120 attitude flight control system was known to have a low-bandwidth control response, so this result was not unexpected. With this limitation in mind, the MTE was successful in exposing the handling qualities deficiencies associated with the aircraft controller. More work is needed with other fixed-wing vehicles to properly size inputs and refine performance requirements, but this MTE shows promise for inclusion in a UAS handling qualities specification.

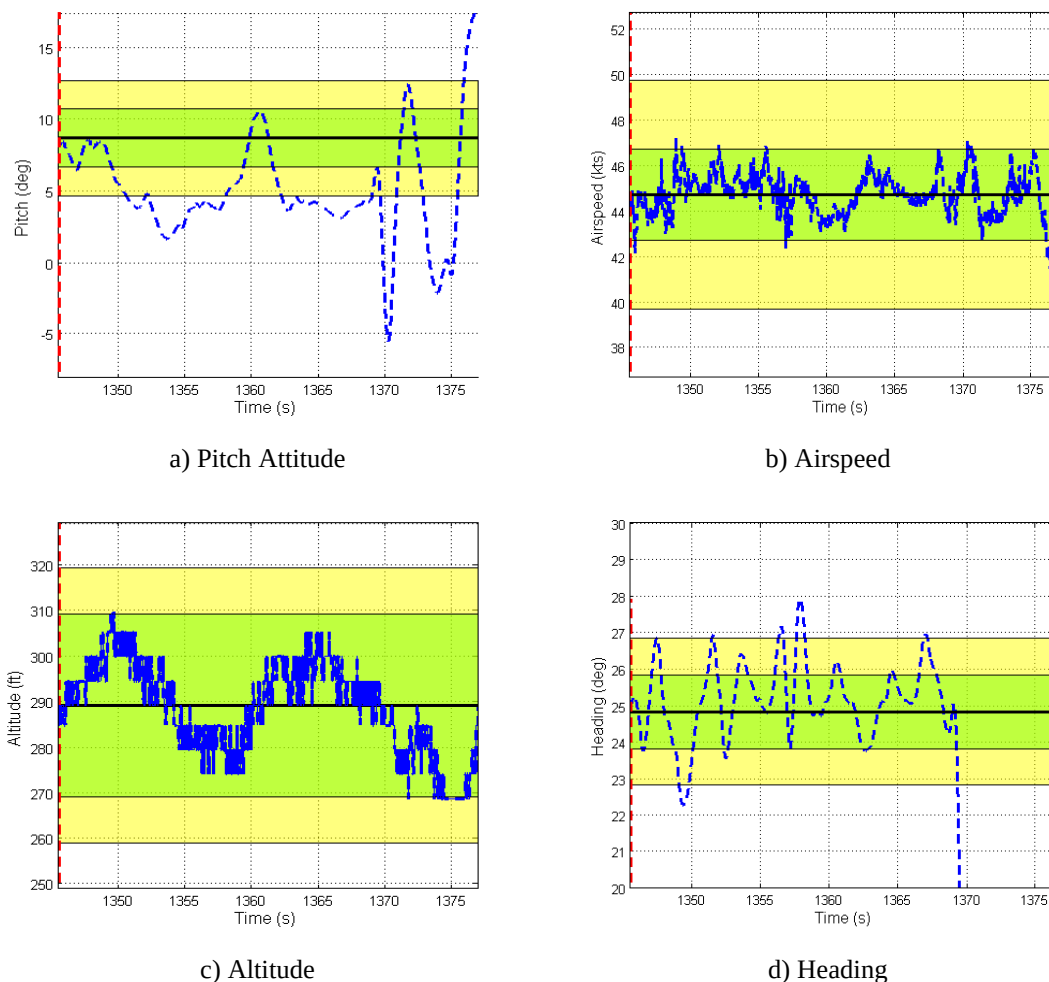


Fig. 8: Flight 14 flightpath regulation in the presence of a discrete gust MTE.

2. Flightpath Regulation in the Presence of a Sum-of-Sines Disturbance (Altitude):

Two sum-of-sines disturbance regulation tasks were considered in this program. Both feature the same Fibonacci-based sum-of-sines signal described previously. The first was introduced as an altitude disturbance of ± 10 meters. The aggressiveness of the task can be adjusted by sizing the input. Example flight test results from Flight 14 are shown in Fig. 9. Because of the flight test course visual line of sight limitations, many of the runs were cut short due to the need to remain within the test area (see Fig. 9d). Before the turn was initiated, the vehicle was able to maintain altitude and airspeed within (mostly) the desired performance bounds. There were more issues with the pitch attitude performance; however, the current requirements on this parameter may be too stringent for a non-precision, non-aggressive task. Again, more work is needed with other fixed-wing vehicles to properly size inputs and refine performance requirements, but this MTE shows promise for inclusion in a UAS handling qualities specification. With a more consistent MTE set (i.e., fewer heading changes while responding to the disturbance), a complete controller-vehicle system analysis can be conducted because of the known disturbance input.

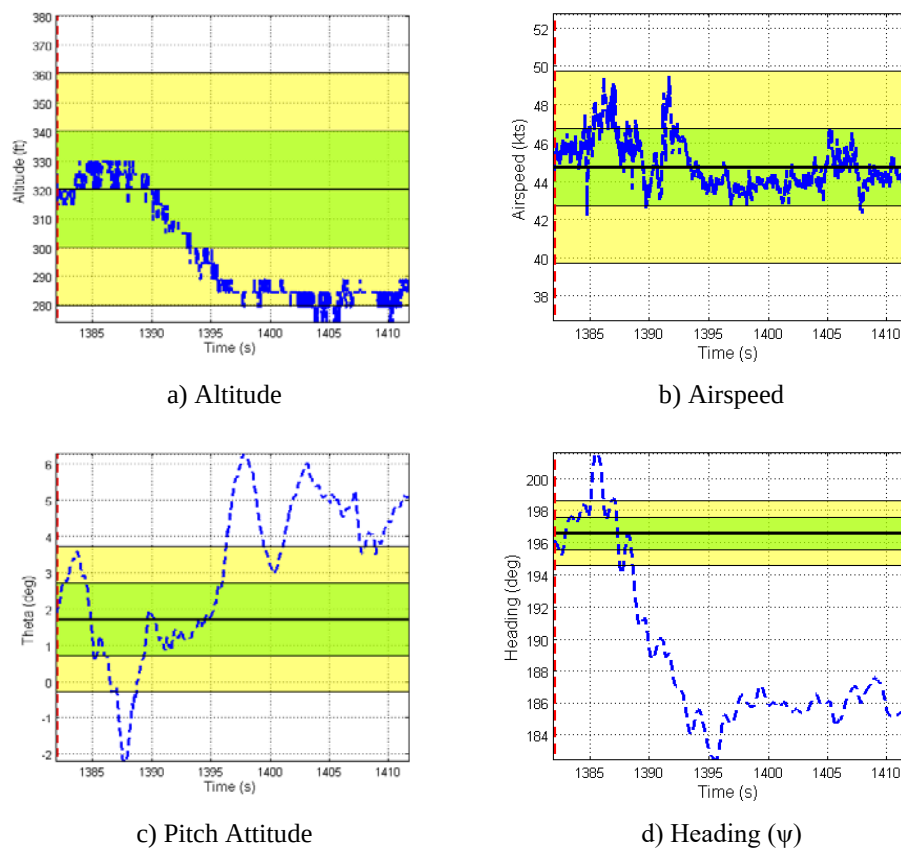


Fig. 9: Flight 14 flightpath regulation in the presence of a sum-of-sines altitude disturbance MTE.

3. Sum-of-Sines Attitude Disturbance:

The attitude sum-of-sines disturbance signal, seen in Fig. 10a, is added to the reference signal of the attitude controller. This task was also flown in a “pseudo” autonomous mode. Unlike the altitude disturbance task, this MTE is intended to be a precision, non-aggressive task. For the example case shown here, the aircraft is flying straight and level for the beginning portions of the maneuver. Then the aircraft banks, e.g., see Fig. 10b, to initiate a turn to remain within the desired test area. MTE performance plots are shown in Fig. 11. During the section of the maneuver where the aircraft is straight and level, the altitude performance generally meets desired performance requirements, while airspeed meets adequate performance requirements. Because of the turn, it is difficult to assess the key performance indicator, pitch attitude, but it does seem likely that at least adequate performance may have been attainable. As with the previous examples, this fixed-wing MTE shows promise, but further refinements are likely needed before it is ready for inclusion in the UAS handling qualities specification.

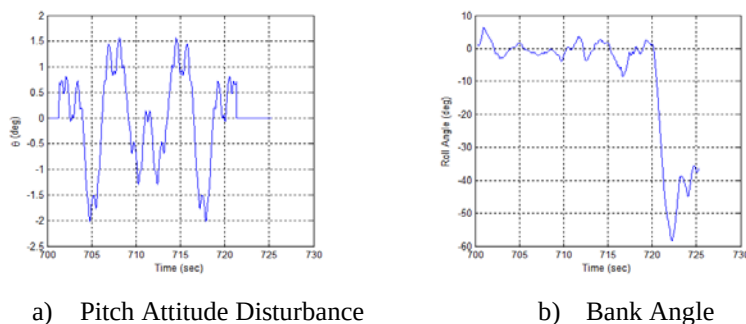


Fig. 10: Flight 16 example sum-of-sines attitude disturbance.

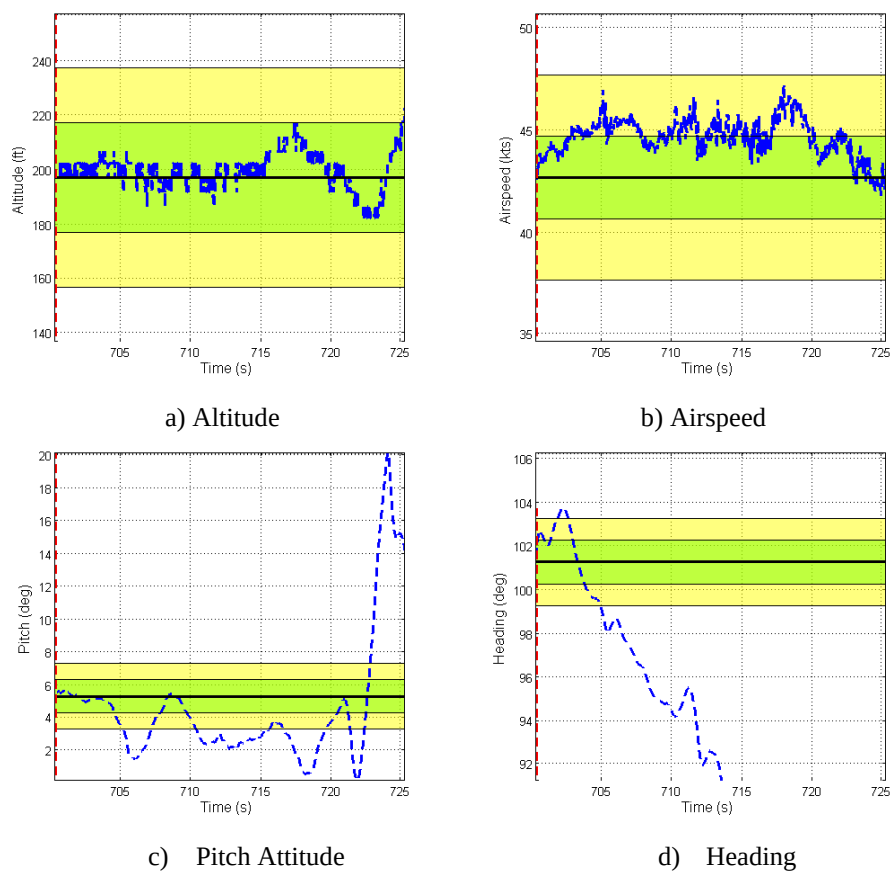


Fig. 11: Flight 16 flightpath regulation in the presence of a SOS attitude disturbance MTE, no delay.

B. Waypoint Following

The Waypoint Following task was flown as a completely autonomous MTE. The MTE featured a square x-y course with altitude changes. Example x-y trajectories are shown in Fig. 12 for the Nominal (blue trajectory) and Added Delay (red trajectory) configurations. Note that there is little difference between the two cases. Furthermore, the vehicle does not meet even a generous set of performance requirements, ± 50 ft/ ± 100 ft, although in this case, the waypoints are generally within adequate bounds. No refinements to the controller were made to improve performance and, as discussed earlier, there were known limitations. An important question regarding the performance requirements will ultimately determine precision and aggressiveness levels. Specifically, is the vehicle required to meet the performance requirements throughout its trajectory or only in the vicinity of the waypoints? Despite these questions, this MTE also shows great promise. With refinements, this MTE will likely be a good discriminator of UAS handling qualities.

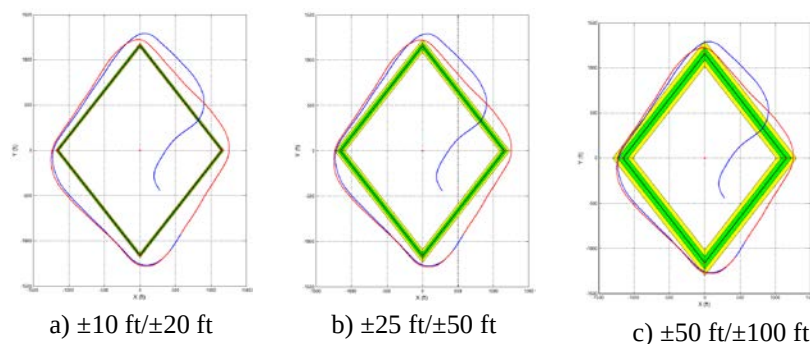


Fig. 12: Flight 16 waypoint following

C. Precision Lateral Offset Landing

1. Line of Sight MTE

As defined previously, the Precision Lateral Offset Landing task was executed as a pilot's line-of-sight task. Fig. 13 shows all Flight 16 runs of the offset landing. In total, there were 4 offset landing attempts flown. The majority of the runs missed the adequate box, where the center of the desired and adequate boxes is 50 ft to the left and 10 ft back of the operator's position (runway). The centerline of the adequate and desired boxes was marked with one or two cones in the middle of the field. The Pilot noted that it was difficult to setup the offset landing task with the snow bank and 24 inches of snow on the ground. Furthermore, the pilot also mentioned that he did not have a good sight line to establish an initial approach. He also noted that the case with 160 ms of added delay was flyable, and the landing task was feasible with this delay present.

The results from all four MTE evaluation flights showed similar results. That is, without adequate cueing present, and this may not be feasible to consistently attain in a line of sight task, the task performance was effectively random. A "good" handling configuration may miss the mark, while a "poor" handling configuration ends up in the box. Thus, when flown in this manner, the MTE will not properly expose UAS handling qualities. There may still be value in an offset landing task if an aircraft point-of-view display is available with proper cueing, such as was used in the AirSTARS flight test program. This is discussed further below.

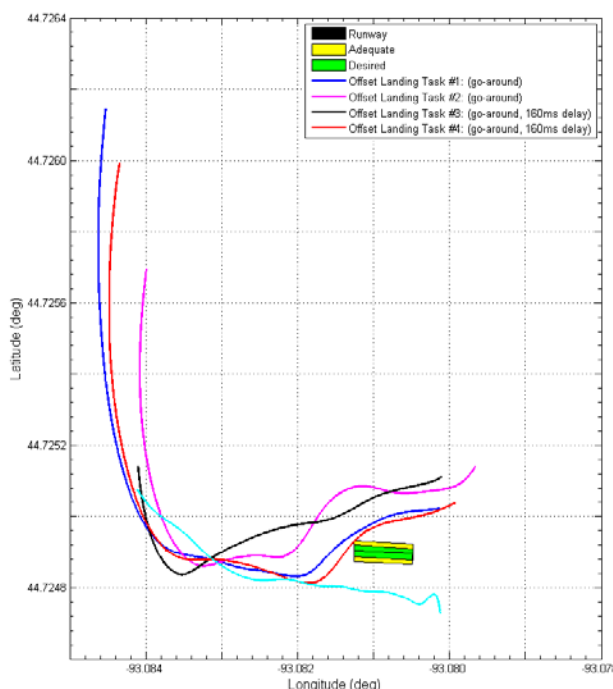


Fig. 13: Flight 16 precision lateral offset landing ground tracks.

2. Point of View MTE

While the UltraStick120 test results with the precision lateral offset landing task clearly exposed the limitations of the MTE as a visual line-of-sight task, previous work at NASA Langley with the AirSTARS vehicle was more successful. In this case, the task was flown as pilot point-of-view (POV) task from a ground station using a synthetic head-up display. In this application, the MTE was successfully executed and did expose handling qualities variations between aircraft configurations.

The maneuver was conducted at a nominal altitude of 1000 feet to avoid any damage to the aircraft and provide the safety pilot with ample window to recover the aircraft from any unexpected situation (e.g., a loss of control departure). The runway was 'simulated' 100 feet below the aircraft, at a longitudinal distance of 1800 feet and a lateral offset of 100 feet towards starboard side of the aircraft. The simulated landing was carried out by projecting a virtual runway on the head-up display (HUD) of the research pilot. A screenshot of the HUD, taken from Ref. 18, is provided in Fig. 14 for reference.

The task was to land within a target touchdown reference location on the runway after performing an ‘S’ maneuver to align with the runway centerline. Performance of the pilot-vehicle system was evaluated using a set of desired and adequate performance requirements.

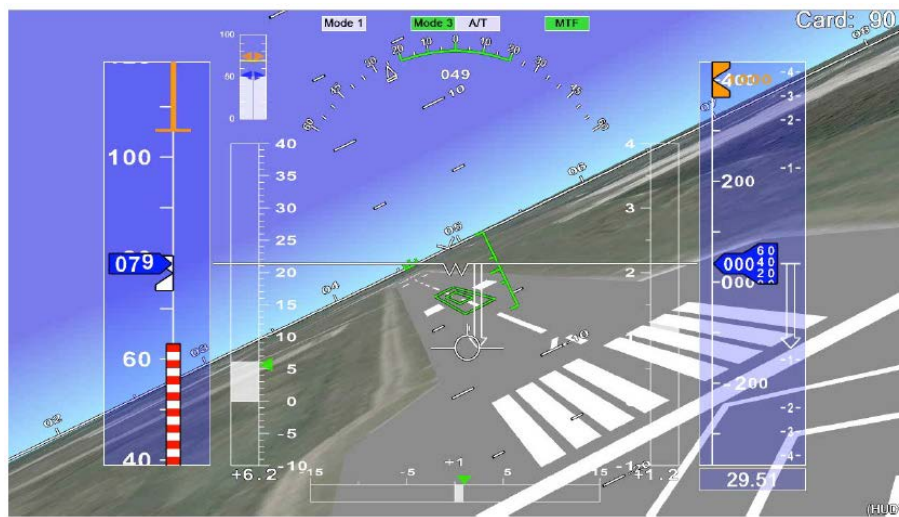


Fig. 14: Head-Up Display (HUD) for the research pilot depicting virtual runway (Ref. 18).

The desired and adequate performance requirements for the offset landing task are provided in Ref. 18 and are reproduced in Table 5 for convenience.

Table 5: Performance Requirements for Offset Landing Task (Ref. 18)

Parameter	Target	Desired	Adequate
Bank Angle (deg)	0	$\pm 10^\circ$	$\pm 20^\circ$
Flightpath Angle (deg)	-3	$\pm 1^\circ$	$\pm 3^\circ$
Lateral Offset (feet)	0	± 12	± 24
Longitudinal Offset (feet)	0	± 164	± 363

The task is to touchdown within a pre-defined target touchdown zone. To begin the maneuver, the UAS is located at a lateral offset position with respect to the runway. To meet performance requirements, the touchdown should occur with bank angle, longitudinal offset, and lateral offset within the bounds specified in Table 5. The bank angle in mode 1 for runs 1-3 is shown in Fig. 15. The run-to-run signals were adjusted to be plotted on the same time axis. Since all the three runs last between 13.86s and 14.2s, the signals were shortened to the same duration without significant loss of data.

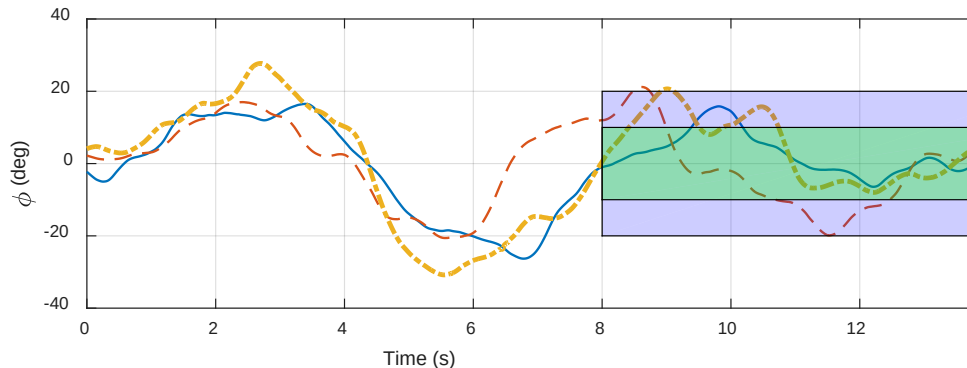


Fig. 15: Bank angle performance for example AirSTARS offset landing runs.

From Fig. 15, the bank angle performance is largely in the adequate range across the entire time interval, with run 3 showing the largest overshoots beyond the desired bounds. It should be noted that since an ‘S’ maneuver is

required to align the aircraft with the runway centerline, the bank angle requirement is applied from the centerline capture to touchdown. Beyond 12s, all three runs have bank angles that fall within the desired range.

Next, the longitudinal and lateral offset performance is considered. All of the runs start with the virtual target touchdown point located at a lateral right shift of 100 feet and 1800 feet in front of the aircraft. Lateral and longitudinal offset specifications are provided to illustrate the performance at touchdown. The objective is to land as close to the reference touchdown point as possible. Fig. 16 shows the lateral and longitudinal performance of the aircraft for runs 1-3 during the offset landing maneuver.

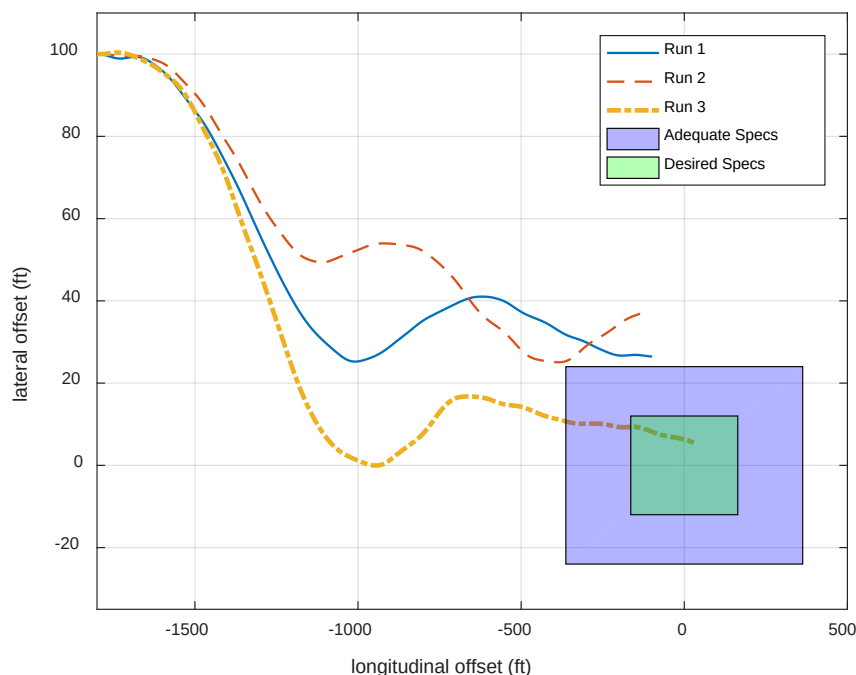


Fig. 16: Lateral and longitudinal offset performance- runs 1-3, flight 44, card 86.

The position of the aircraft with respect to the runway is not directly available as a data signal. Therefore, the lateral and longitudinal positions were computed using heading angle data and airspeed data available during the run. From heading angle and airspeed, lateral and longitudinal components of aircraft velocity are found. Positions can then be determined by integrating the velocities in a discrete manner. Fig. 16 shows that of the three runs, only run 3 meets the performance requirements; in fact, run 3 meets the desired specifications very well. Run 2 is possibly the most interesting in this regard, since the aircraft goes off-course just as it approaches the target touchdown area. In general, the AirSTARS results indicate a POV Precision Lateral Offset Landing MTE can be an appropriate discriminator of UAS handling qualities.

IV. Conclusions and Lessons Learned

The application of traditional aircraft requirements and classification schemes will prove challenging for the new and varied classes of unmanned aerial systems coming to the airspace. The mission-oriented approach, with associated mission task elements, is uniquely suited to address the certification requirements of these new vehicles, designs, and missions. A template for the mission task elements has been defined, based upon the format utilized for the rotorcraft design specification. To demonstrate the ability of these mission task elements to distinguish between handling qualities levels, four mission task elements were developed and flown for a fixed wing UAS. The following observations were made based upon this testing:

- The mission task elements under investigation demonstrated the ability or promise to be good discriminators of handling qualities.

- Data are of paramount importance. One vehicle flying one set of tasks does not define the final limits and performance bounds; it can simply add to the database and contribute to the proof that the task is valid.
- Particularly during the early mission task element development phases, complete knowledge of the vehicle and its capabilities is critical.
- Mission task elements must be designed for the mission and its requirements, not for the performance of any one test vehicle. The vehicle will, of course, influence the requirements and perhaps mission design. However, the mission task elements should align with the mission and vehicles capable of performing that mission.

Lessons learned:

- Where possible, do not introduce additional vehicle commands when not a part of the task, i.e., avoid turning while flying a flightpath regulation task that does not include a turning element.
- Be mindful of the limitations of the task and cueing. The offset landing task performed with direct visuals from the ground offers different, and perhaps unnecessary, challenges that a head-up-display indicating a vehicle-centered flightpath view may otherwise mitigate.
- Be clear about the task and its wording. Do the performance requirements apply during the entirety of the task or only at certain locations? If only at certain locations, what is considered desired/acceptable behavior in the other portions of the task?
- Recognize that the same mission task element may be suitable for addressing and evaluating different missions with different requirements.

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